MASKSIM: FREQUENCY-DOMAIN DETECTION OF AI-GENERATED IMAGES

Project Synopsis – Final Evaluation

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1. Abstract

MaskSim is an end-to-end system that decides whether a photograph was produced by a camera or synthesised by a generative model, and — when it was synthesised — which generator produced it. The method is a signal-processing one rather than a pattern-recognition one: every image generator ends by upsampling a small latent representation with learned filters, and a finite-length learned filter cannot fully suppress the spectral replicas that upsampling creates. The result is a faint periodic residue, below one grey level in 255, located at frequencies fixed by the network weights rather than by the picture. The system removes image content with a pre-trained DnCNN denoiser, takes the log-magnitude Discrete Fourier Transform of the residual, multiplies it by a learned per-frequency mask, and correlates the result against a learned reference pattern. Seven per-generator detectors are combined by a two-stage calibrated maximum, and the output is reported as a false-positive rate measured against photographs known to be real, not as a binary verdict. The complete application integrates a FastAPI backend, PyTorch inference services and a browser front end that exposes every intermediate tensor, together with an interactive explainer that re-implements the entire method in JavaScript and trains a miniature detector live in the browser.

2. Problem Statement and Objectives

Synthetic images are no longer separable from photographs by eye, and detectors built on semantic cues — malformed hands, incorrect text, inconsistent reflections — are obsoleted with each model release. Metadata and watermarks are trivially stripped and are absent from open-weight models. A detector must therefore measure a property of the *manufacturing process* rather than of the depicted content, and must state its uncertainty in a form an operator can act on.

- Detect the periodic artifact left by a generator's upsampling stack, rather than image semantics.
- Localise the artifact in the frequency domain, where a periodic signal is concentrated rather than diffuse.
- Learn one weight per frequency bin, so that the trained model can be inspected and explained.
- Report a calibrated false-positive rate instead of a probability or a fixed threshold.
- Apply identical preprocessing during training and inference, and never resample the input image.
- Expose the pipeline through a web application that shows every intermediate stage, and through an interactive explainer that reconstructs the method from first principles.

3. Overall System Architecture

The application follows the pipeline: User Image → Browser Front End → FastAPI /api/detect → Centre Crop and Colour Transform → DnCNN Residual → log |DFT| → 1×1 Channel Mixer → Learned Mask → BatchNorm → Cosine Similarity against Reference → Logistic → Calibration Table → False-Positive Rate → JSON Response → Front-End Result and Frequency Maps.

- Frontend: hand-written HTML, CSS and JavaScript with no external libraries; upload, verdict ring, stage strip, per-detector frequency explorer and a radix-2 FFT implemented in-browser.
- Backend: FastAPI served with Uvicorn; all seven checkpoints are loaded once at startup and cached for subsequent requests.
- ML: PyTorch Lightning MaskSim models with the published champion checkpoints (≈23 MB each).
- Shared state: the most recently analysed image is held in memory so that the detection page and the explainer page operate on the same input.
- Detector routing: Stable Diffusion 1→D1, Stable Diffusion 2→D2, Stable Diffusion XL→D3, DALL·E 2→D4, DALL·E 3→D5, Midjourney→D6, Firefly→D7; the generic model combines all seven.

4. DSP Preprocessing Pipeline

Every training and inference image follows the same deterministic sequence. A fixed 512×512 centre crop is taken with its origin snapped down to a multiple of eight, so that the crop aligns with the 8×8 block grid JPEG quantises on and the resulting blocking artifact lands on the same frequency bins in every image. The crop is converted from RGB to YCbCr — the basis JPEG and WebP quantise in — and scaled to the range [0,1]. A pre-trained 17-layer DnCNN is then applied and its output, which is an estimate of the *noise* rather than of the clean image, is retained; the scene itself is discarded. Because a 17-layer 3×3 network has a 17-pixel invalid border, 17 pixels are removed from each edge and the residual is 478×478. The two-dimensional DFT of the residual is taken, the magnitude is retained and the phase discarded, and the logarithm is applied.

Each of those decisions follows from a property of the transform. Discarding phase buys exact invariance to the crop position, because translating an image multiplies every coefficient by a unit-magnitude complex number and leaves the magnitude unchanged. The logarithm compresses six decades of dynamic range and, more importantly, converts the generator's multiplicative gain into an additive offset, which is what makes a linear per-bin comparison between the two classes meaningful. The refusal to resize follows from the scaling theorem: resampling by a factor s moves a component at bin k to bin k/s, so a rescaled image no longer aligns with the mask. Preprocessing consistency between training and inference is therefore part of the model deployment contract.

5. The Detection Model and Training

The deployed system contains seven per-generator detectors sharing one frozen denoiser. Unlike a conventional classifier, the trainable model contains no convolutional feature extractor: it is two maps of numbers, one entry per frequency bin, plus seventeen further parameters.

Stage	Operation	Shape	Parameters	Trained
S1	Centre crop, 8-aligned, RGB → YCbCr	3 × 512 × 512	0	No
S2	DnCNN residual (17 layers, 64 ch)	→ 3 × 478 × 478	≈558,000	No — frozen
S3	log |FFT2|, DC centred	3 × 478 × 478	0	No
S4	1×1 convolution, channel mixing	bin-wise	9	Yes
S5	Mask M = σ(boxblur(θ))	3 × 478 × 478	685,926	Yes
S6	BatchNorm, per channel	3 × 228,484	6	Yes
S7	Cosine similarity with reference R	3 × 478 × 478	685,926	Yes
S8	Logistic p = σ(e^a·cos + b)	scalar	2	Yes
S9	Calibration → false-positive rate	scalar	0	Measured

Total trainable parameters are 1,371,869, of which 1,371,852 are the mask and the reference. The mask is parameterised as a sigmoid of a fixed 3×3 box-blurred free tensor initialised from $N(0, 0.01^2)$, so the mask begins at approximately 0.5 everywhere — no preference for any frequency — and the box blur acts as a smoothing prior that costs no parameters. Training uses Adam at a learning rate of 1×10^{-2} for the head, approximately twenty epochs, with JPEG re-encoding at a random quality between 65 and 100 applied afresh every epoch; the per-epoch randomness is the principal regulariser. Training takes approximately three hours per detector on a single GPU and is bound by disk I/O and the frozen denoiser rather than by the learned arithmetic.

The loss contains one deliberate modification that governs the entire result. Synthetic images are driven towards a cosine of +1, but pristine images are driven towards $|cos| = 0$ rather than towards −1. Requiring anti-correlation would reward the model for constructing a template of what real images look like, which is how a generator detector silently becomes a dataset classifier. Requiring orthogonality states correctly that a real image should have no projection onto the generator's template in either direction.

6. Dataset and Data Integrity

- Synthetic images are drawn from Synthbuster, in which the same prompts and scenes are rendered by each of the seven generators, so the two classes differ by generator rather than by subject matter.
- Pristine images are camera photographs from RAISE, COCO and FlickrFaces; the same pool supplies both training negatives and the calibration tables.
- Approximately 12,000 real photographs per detector form the calibration tables, stored as CSV files; a further table of maxima supports the generic model.
- Synthetic images must be unmodified generator output — never re-saved and never resized — because both operations destroy the artifact being measured.
- Class balance and source matching are treated as integrity requirements, not conveniences: if the two folders differ in camera pipeline or compression history, a naive model achieves high accuracy on its own split and fails on every other source.
- The total preparation set, obtained through the repository's data_preparation.sh, is approximately 30 GB.

7. Results and Verification

The reference implementation was executed successfully end to end. All seven checkpoints load, produce scores for a supplied test image, and the generic model returns a calibrated false-positive rate; the web application was

verified to reproduce the same numbers as the command-line tool, because it calls the repository's own functions rather than re-implementing them.

A complete miniature re-implementation of the method was additionally written from scratch in approximately 400 lines of JavaScript and is trained live in the browser on 28 controlled images (14 clean, 14 carrying a planted five-frequency artifact at 1.5/255), with a calibration set of 60 held-out images. The fitted mask opened 65 bins out of 65,536. Measured results on four conditions were as follows.

Condition	Artifact present	Measured FPR	Verdict band
Source image, untouched	No	81.7 %	No evidence
Artifact planted at 1.5/255	Yes	0.0 %	Very likely synthetic
Planted, then JPEG quality 60	Yes	0.0 %	Very likely synthetic
Planted, then resized to 125 %	Yes	28.3 %	No evidence

The fourth row is the principal experimental finding and it confirms the scaling theorem quantitatively: a resize of twenty-five per cent returns a confidently detected image to the population of ordinary photographs, because the displacement $\Delta k \approx k(s-1)$ exceeds the width of the mask's open regions. Two further controlled observations were recorded. Warm-starting the mask from its closed-form optimum degraded validation AUC from 0.89 to 0.69, because a near-zero mask causes the following normalisation to amplify noise; and precomputing the augmented spectra rather than re-randomising them each epoch degraded AUC from 0.89 to 0.63, confirming that the augmentation is the regulariser.

A measured engineering result: because all seven detectors share one frozen denoiser, computing the residual once instead of fourteen times reduces CPU inference from approximately one minute to approximately five seconds per image. The denoiser accounts for about 1.46×10^{11} multiply-accumulates per image, against roughly 1.4×10^{6} for the entire learned head.

Published cross-generator transfer figures from the source paper are reported by the authors and were not re-measured in this project; a detector trained on one generator transfers well to models sharing a decoder architecture and poorly otherwise. An independent quantitative evaluation on a held-out real-image test set is listed in Section 10 as outstanding work.

8. Backend, Frontend and Application Integration

The FastAPI backend loads all seven neural models at startup rather than per request, and at startup fingerprints each checkpoint's denoiser weights; only if all seven carry an identical frozen denoiser does it enable the shared-residual optimisation, otherwise it falls back to the slow, always-correct path. The API exposes upload and analysis, sample scoring, and a shared-source endpoint that returns immediately with a thumbnail and scores the image on a background thread, which the front end polls.

The front end communicates using multipart/form-data. Result rendering handles synthetic matches, negative results, images below the 512×512 minimum, and stale-server detection through a build string. Score conversion is performed exactly once: backend scores in the 0–1 range are displayed as percentages in the interface. Every result is accompanied by the four real pipeline stages, a similarity map, the trained mask and the masked spectrum for each of the seven detectors.

A second page reconstructs the method interactively on the user's own uploaded image, with a hand-written radix-2 FFT, a probe that draws the single stripe pattern any selected frequency bin represents at its true amplitude and phase, and the miniature detector described in Section 7. Both pages operate on the same shared image, so an upload on either is analysed by both.

9. Validation and Current Status

- Reference pipeline: PASS; all seven checkpoints load and score a supplied image, and the generic model returns a calibrated FPR.
- Wrapper equivalence: PASS; the web application calls the repository's own preprocessing, scoring, similarity-map and calibration routines, so it agrees with the command-line tool by construction.
- API: PASS for upload/analyse, sample scoring, shared-source upload, background scoring and source reset; verified against a stubbed engine including the below-minimum-size rejection path.
- Client-side re-implementation: PASS; the hand-written FFT round-trips correctly and the miniature detector trains, calibrates and scores without error.
- Browser validation: PASS; all seven pipeline stages, the frequency probe, the blind test and the eight-stage walkthrough render with no JavaScript errors.
- Shared-denoiser optimisation: PASS; guarded by a weight fingerprint with automatic fallback.
- Outstanding: an independent quantitative evaluation (ROC, AUC, cross-generator matrix) on a held-out test set has not yet been performed and is not claimed.
- Current project status: OPERATIONAL; reproduction and application complete, quantitative evaluation pending.

10. Critical Points and Future Improvements

The principal technical limitation is structural rather than implementational. Because the mask examines fixed frequency coordinates, any operation that changes the sampling grid displaces the artifact while the detector continues to examine the original bins. A resize of two to three per cent halves the available evidence, and almost every image that has passed through a social network has been resized. A second limitation is that the method fails silently: an image from a generator none of the seven detectors was trained on scores like a photograph, which is the safe direction to fail in but gives no indication that the detector is outside its competence. A negative result is therefore evidence of the absence of seven specific fingerprints and is not evidence of authenticity.

- Perform a quantitative evaluation with ROC curves, AUC and a 7×7 cross-generator matrix on a held-out test set.
- Measure robustness explicitly: AUC as a function of JPEG quality and of resize factor, to test the scaling theorem directly.
- Run ablations removing the denoiser, opening the mask fully, and substituting linear magnitude for log magnitude.
- Compare against a simple baseline — logistic regression on the radial energy profile alone — to quantify what the learned mask contributes.
- Implement periodic-plus-smooth decomposition, which the source paper cites and does not apply, and measure whether removing the DFT wrap-around artifact improves separation.
- Extend the method to a patch-wise variant so that partially inpainted images, currently out of scope, can be handled.
- Investigate the same derivation for synthetic audio: neural vocoders upsample a mel-spectrogram to the sample rate by a factor of approximately 256 using transposed convolutions, producing the identical multirate structure at fixed frequencies in hertz.

11. Conclusion

MaskSim demonstrates a complete detection pipeline grounded in multirate signal processing rather than in learned image semantics. It combines deterministic DSP preprocessing, a frozen denoiser used as a content suppressor, a Fourier-domain matched filter whose weights are a single number per frequency bin, and a calibration stage that converts an uninterpretable score into an actionable false-alarm rate. The reproduction executes end to end against the published checkpoints, the web application exposes every intermediate tensor rather than only a verdict, and an independent re-implementation of the method was written from scratch and verified on controlled data, where it reproduced both the expected detection behaviour and the expected failure under resampling. The resulting system covers the full seven-generator output space, documents each design decision by the transform property that justifies it, and states its own limitations in the form in which they follow from the mathematics.

12. References:

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